

Replication Package

“Superstars or Supervillains? Large Firms in the South Korean Growth Miracle”

September 2026

This note explains the replication package for the paper “Superstars or Supervillains? Large Firms in the South Korean Growth Miracle.” Table 1 summarizes which codes generate the outputs of the draft.

1. REPLICATION STEPS

To replicate the package, the procedure is as follows:

1. Run the do-files in Stata Part 1, located in the STATA folder. After setting the directory in SHELL.do, run the following codes in order:
 - (a) Run STATA/DATA_CLEANING.do. This code cleans the dataset used for computing firm- and sector-level datasets.
 - (b) Run STATA/CONCENTRATION.do. This code computes the concentration figures.
 - (c) Run STATA/TO_MATLAB.do. This code creates inputs for quantitative exercises.
 - (d) Run STATA/ESTIMATION_PROD_FUNC.do. This code creates data inputs for the production-function estimation procedure.
2. Run the Matlab file MATLAB_PROD_FUNC/SHELL.m, which runs MATLAB_PROD_FUNC/DKL_MARKUP.m, MATLAB_PROD_FUNC/PROD_FUNC_MARKUP.m, and MATLAB_PROD_FUNC/PROD_FUNC_BW_MARKUP.m consecutively.
 - (a) MATLAB_PROD_FUNC/DKL_MARKUP.m estimates DKL markups.
 - (b) MATLAB_PROD_FUNC/PROD_FUNC_MARKUP.m estimates production functions based on $\sigma = 5$.
 - (c) MATLAB_PROD_FUNC/PROD_FUNC_BW_MARKUP.m estimates production functions based on the elasticity of substitution from Broda and Weinstein (2006, QJE).
3. Run MATLAB/MATLAB_part1.m.
4. Run Stata Part 2. Run the following codes consecutively. Stata Part 2 requires the backed-out firm-level shocks generated by MATLAB/MATLAB_part1.m.
 - (a) Run STATA/PF_EST_RESULT.do. This code reports the production-function estimation results.
 - (b) Run STATA/Merger.do. This code runs robustness checks regarding mergers.
 - (c) Run STATA/SHOCK_CORR.do. This code runs robustness checks regarding shock correlations.
 - (d) Run STATA/DISTRIBUTED_LAG.do. This code estimates the distributed-lag model between firm-level shocks and observables.
 - (e) Run STATA/spillover_region.do. This code constructs a shock series that incorporates potential spatial spillovers, estimated from Choi and Shim (REStat, forthcoming).

- (f) Run STATA/HCI_Drive.do. This code runs the HCI Drive industrial-policy event study.
 - (g) Run STATA/BRIBERY.do. This code runs external validation regarding bribery.
 - (h) Run STATA/SHOCK_VALIDATION.do. This code runs additional robustness checks that validate the shocks.
 - (i) Run STATA/FIGURE6_scatter_plot_villain.do. This code plots Figure 6.
5. Run MATLAB/MATLAB_part2.m.

2. MATLAB CODES

Below, we explain the m-files called by MATLAB/MATLAB_part1.m and MATLAB/MATLAB_part2.m. All m-files are located in the MATLAB folder.

1. MATLAB/MATLAB_part1.m runs the following codes in order:
 - (a) save_result.m: This code backs out firm-level productivity and distortions using Proposition 2.1 and writes the firm-level shocks used in Stata Part 2:
 - OUTPUT/result_base.csv
 - OUTPUT/result_base_notruncation_nosmoothing.csv
 - OUTPUT/result_counterfactual_a.csv
 - (b) solve_full_model_part1.m. This code solves the baseline model and runs the main counterfactuals.
 - i. It backs out firm-level and aggregate shocks by matching the data (model_extract_shock.m and model_extract_shock_dynamics.m) and summarizes them in collect_results1.m. This part produces Figures 2, 3, and B5 and Table 2.
 - ii. removing_granular_shocks.m replaces the top-3 firms' shocks with the sectoral averages. This part produces Figures 4 and B13.
 - iii. removing_samsung_shocks.m and removing_hyundai_shocks.m run the same counterfactual for Samsung Electronics and Hyundai Motors. This part produces Figures 5 and B14 and Table 3.
 - iv. It solves the model under alternative market structures (Table B4 and Figure B9), and computes domestic markups and a counterfactual that assesses the role of import competition (Figure B7).
 - v. It removes the shocks of each top-3 firm one at a time and writes OUTPUT/villain_all.csv and OUTPUT/sale_gdp_top3.csv, which STATA/FIGURE6_scatter_plot_villain.do uses to plot Figure 6.
 - vi. removing_granular_shocks_top4.m and removing_granular_shocks_top10.m run the counterfactual for the top-4 and top-10 firms, and a further counterfactual holds the capital stock at its observed values. These results are saved for Table B11.
 - vii. Finally, it saves MATLAB/result/p1_to_p2_inputs.mat, which MATLAB/MATLAB_part2.m loads.

- (c) `solve_full_model_only_tau_k.m`. This code engineers the observed rise in sales concentration solely through capital distortions, and produces Panels A–B of Figure 7.
 - (d) `solve_full_model_only_tau_l.m`. This code does the same with labor distortions, and produces Panels C–D of Figure 7.
 - (e) `making_table4.m`. This code combines the results of the two distortion-only counterfactuals into Table 4.
 - (f) `run_robustness.m`. This code calls `solve_full_model_robustness.m` for 17 alternative specifications: alternative values of $\sigma, \rho, \eta, \theta, \psi$, and ζ ; sector-specific σ_j from Broda and Weinstein (2006, QJE); production functions estimated on exporters or non-exporters only; constant returns to scale; and love-of-variety. It then runs `making_tableB11.m`, which combines these results with the top-4, top-10, and fixed-capital results from `solve_full_model_part1.m` into Table B11.
2. `MATLAB/MATLAB_part2.m` runs `solve_full_model_part2.m`, which requires the output of Stata Part 2. This code:
- (a) solves the counterfactuals under alternative assumptions on non-top-3 firms (Table 7):
 - Free entry (`solve_counterfactual_dynamics_profit_fix.m`)
 - Constant sectoral productivity (`solve_counterfactual_dynamics_A_j_fix.m`)
 - Structural productivity spillovers (using the output of `STATA/spillover_region.do`)
 - Reduced-form spillovers (using the output of `STATA/SHOCK_CORR.do`)
 - (b) solves the counterfactuals that strip out the components of firm-level shocks attributable to industrial policy and bribery, using the distributed-lag estimates from `STATA/DISTRIBUTED_LAG.do`. This part produces Table 6.
3. The codes above call the following scripts and functions:
- (a) `set_parameters_load_data.m` sets the parameters and loads the data and the production-function estimates.
 - (b) `model_extract_shock.m` and `model_extract_shock_dynamics.m` back out the shocks in the static and dynamic models, respectively.
 - (c) The following codes solve the equilibrium given the shocks, the steady state, and the transition path to the steady state:
 - `model_given_shock.m`
 - `model_given_shock_dynamics.m`
 - `model_solve_ss.m`
 - `model_until_ss.m`
 - (d) `solve_counterfactual*.m` solve the counterfactual economies.
 - (e) `collect_results1.m` and `collect_results_robust.m` compute aggregate and sectoral statistics, and `CR3_decompose.m` computes the concentration-ratio decomposition in Table 2.

Table 1: Codes and Their Outputs

Figures		Tables	
Exhibit	Generated by	Exhibit	Generated by
Figure 1	CONCENTRATION.do	Table 1	Constructed by hand (calibration summary)
Figure 2	making_figure2.m	Table 2	making_table2.m
Figure 3	making_figure3.m	Table 3	making_table3.m
Figure 4	making_figure4.m	Table 4	making_table4.m
Figure 5	making_figure5.m	Table 5	DISTRIBUTED_LAG.do
Figure 6	FIGURE6_scatter_plot_villain.do	Table 6	making_table6.m
Figure 7, Panels A–B	making_figure7_AB.m	Table 7	making_table7.m
Figure 7, Panels C–D	making_figure7_CD.m		
Figure B1	CONCENTRATION.do	Table B1	Constructed by hand (sector classification)
Figure B2, Panels A–B	CONCENTRATION.do	Table B2	PF_EST_RESULT.do
Figure B2, Panels C–D	SHOCK_VALIDATION.do	Table B3	PF_EST_RESULT.do
Figure B3	CONCENTRATION.do	Table B4	making_tableB4.m
Figure B4	CONCENTRATION.do	Table B5	BRIBERY.do
Figure B5	making_figureB5.m	Table B6	SHOCK_VALIDATION.do
Figure B6	Merger.do	Table B7	SHOCK_VALIDATION.do
Figure B7	making_figureB7.m	Table B8	SHOCK_VALIDATION.do
Figure B8	ESTIMATION_PROD_FUNC.do	Table B9	SHOCK_VALIDATION.do
Figure B9	making_figureB9.m	Table B10	SHOCK_CORR.do
Figure B10	HCI_Drive.do	Table B11	making_tableB11.m
Figure B11	BRIBERY.do		
Figure B12	SHOCK_VALIDATION.do		
Figure B13	making_figureB13.m		
Figure B14	making_figureB14.m		